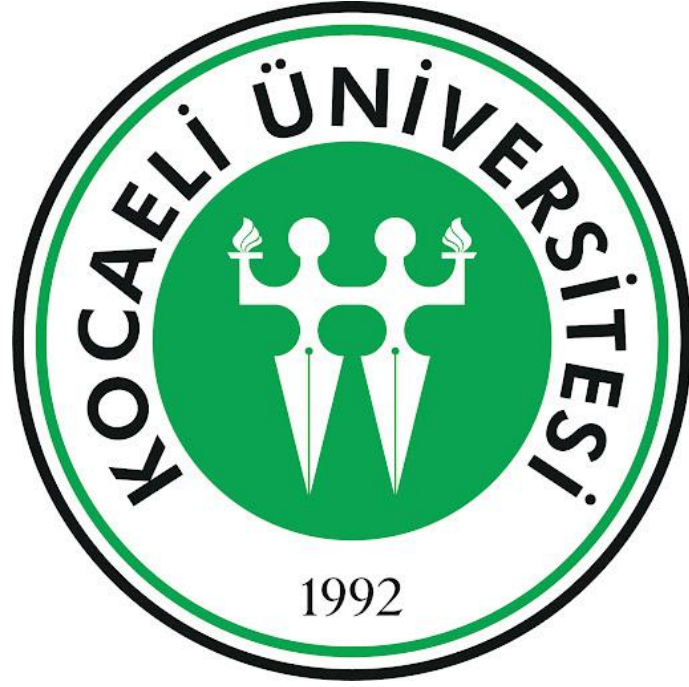




KOCAELİ ÜNİVERSİTESİ

MÜHENDİSLİK FAKÜLTESİ

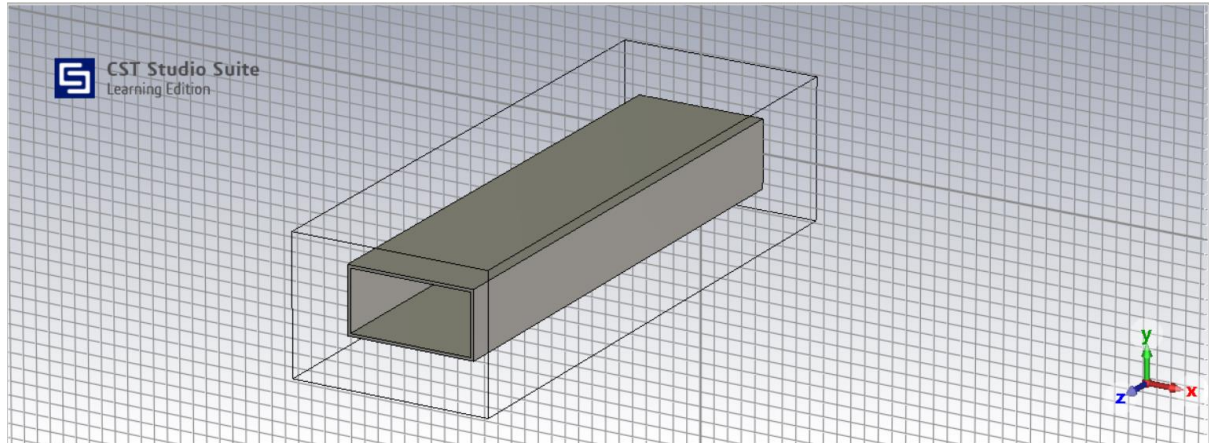
ELEKTRONİK VE HABERLEŞME MÜHENDİSLİĞİ BÖLÜMÜ

CST PROJE RAPORU

230208041-Şevval KARAKURT

Kocaeli-Mayıs 2025

WAVEGUIDE (WR340)



Name:

Xmin: Xmax:

Ymin: Ymax:

Zmin: Zmax:

Component:

Material:

Name:

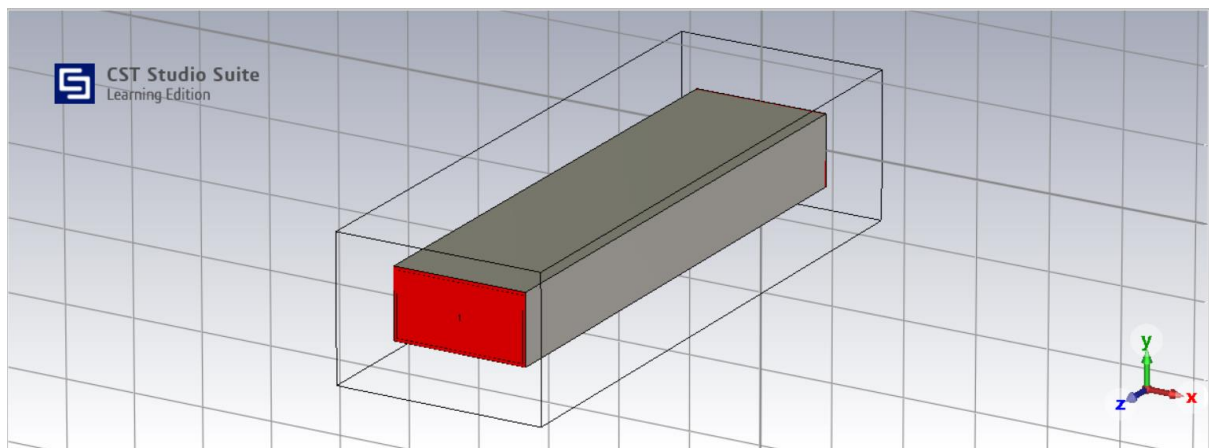
Xmin: Xmax:

Ymin: Ymax:

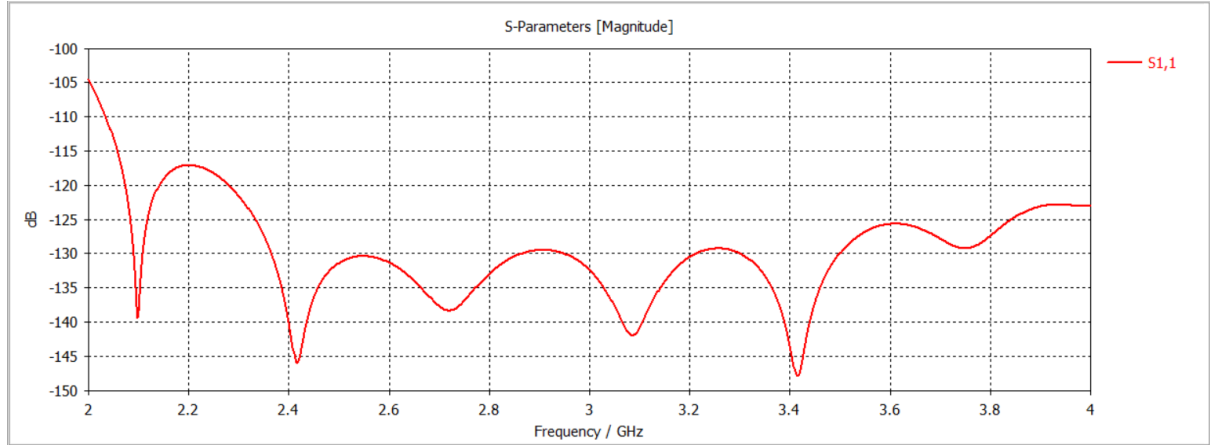
Zmin: Zmax:

Component:

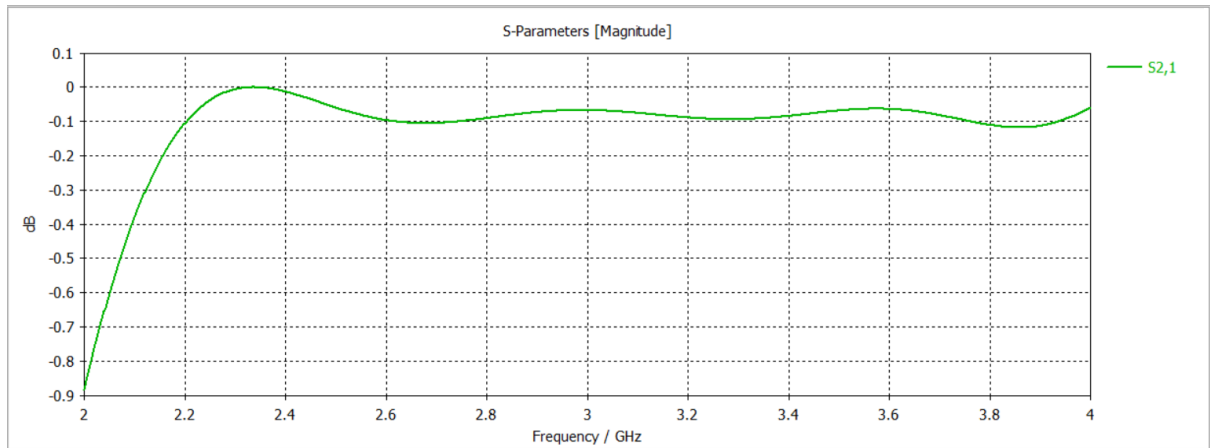
Material:



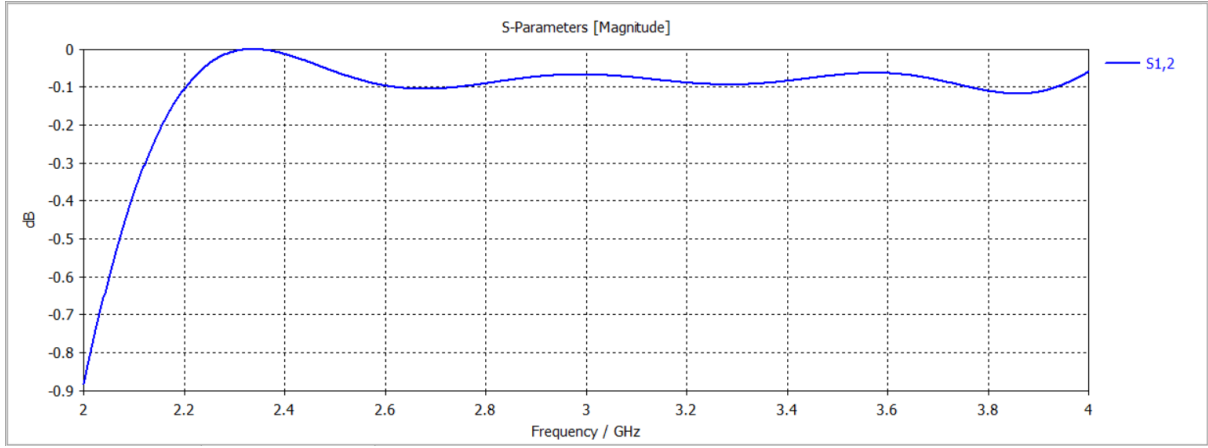
S PARAMETER



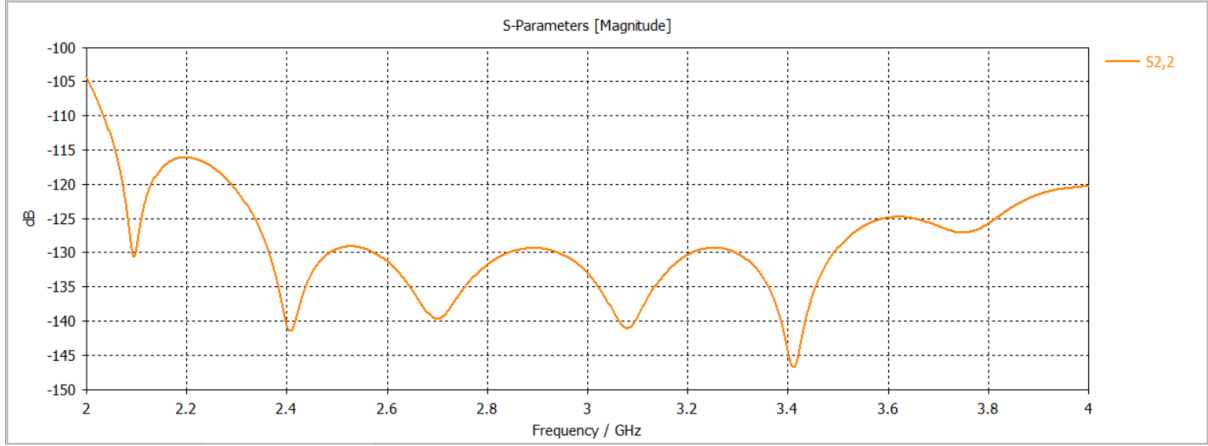
(1 numaralı porta gönderilen bir sinyalin ne kadarının geri yansıdığını gösterir.)



(1 numaralı porttan gönderilen bir sinyalin ne kadarının 2 numaralı porta iletildiğini gösterir.)

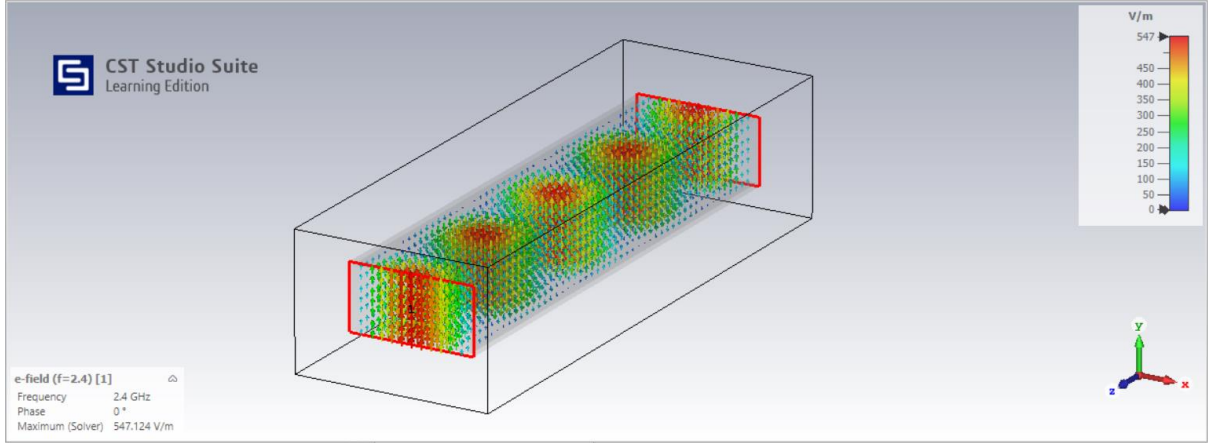


(2 numaralı porttan gönderilen bir sinyalin ne kadarının 1 numaralı porta iletiildiğini gösterir.)

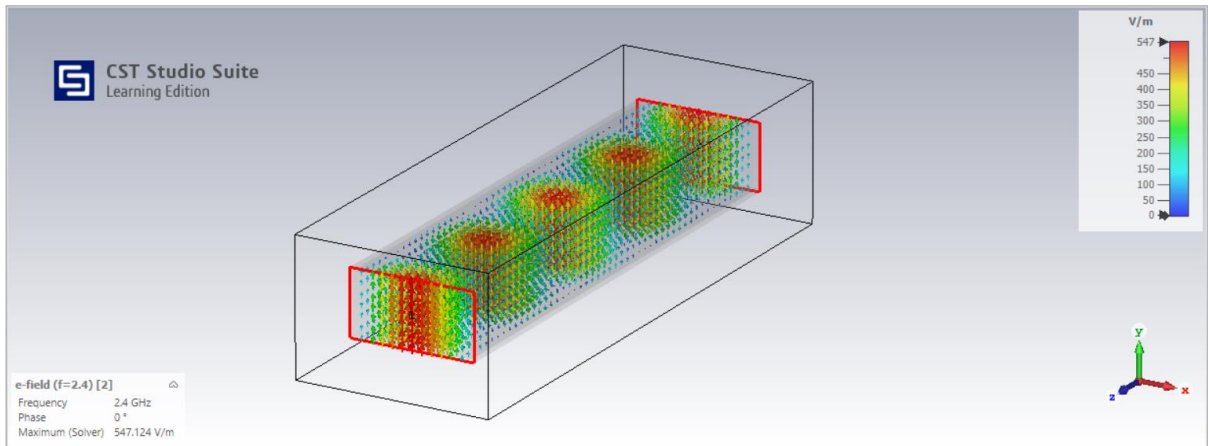


(2 numaralı porta gönderilen bir sinyalin ne kadarının geri yansıdığını gösterir.)

E-FIELD

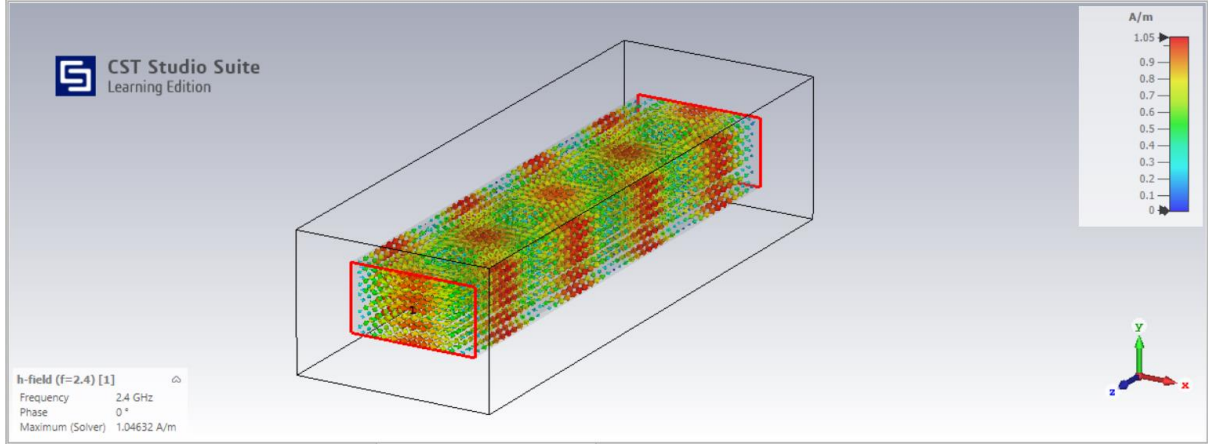


(Giriş portundan (1) gönderilen bir sinyalin dalga kılavuzunda nasıl yayıldığını ve çıkış portuna (2) nasıl ulaştığını gösterir.)

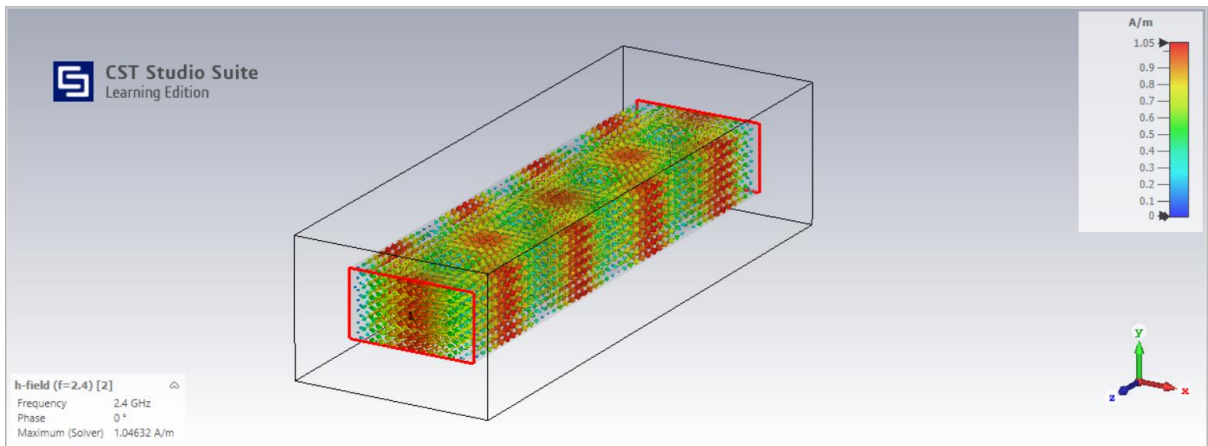


(Çıkış portundan (2) gönderilen bir sinyalin dalga kılavuzunda nasıl yayıldığını ve giriş portuna (1) nasıl ulaştığını gösterir.)

H-FIELD

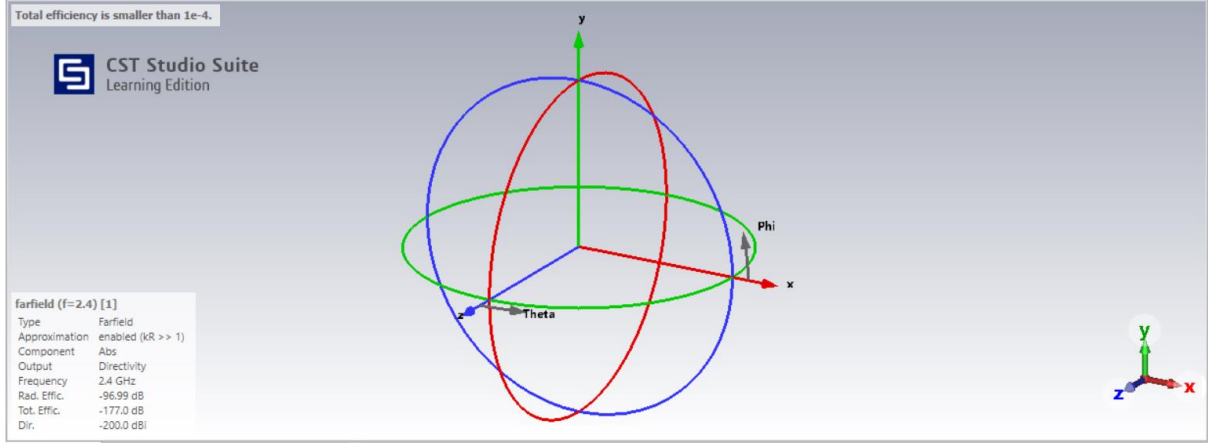


(1 numaralı porttan çıkan ve 2 numaralı porta doğru ilerleyen bir sinyalin oluşturduğu alan dağılımını gösterir.)

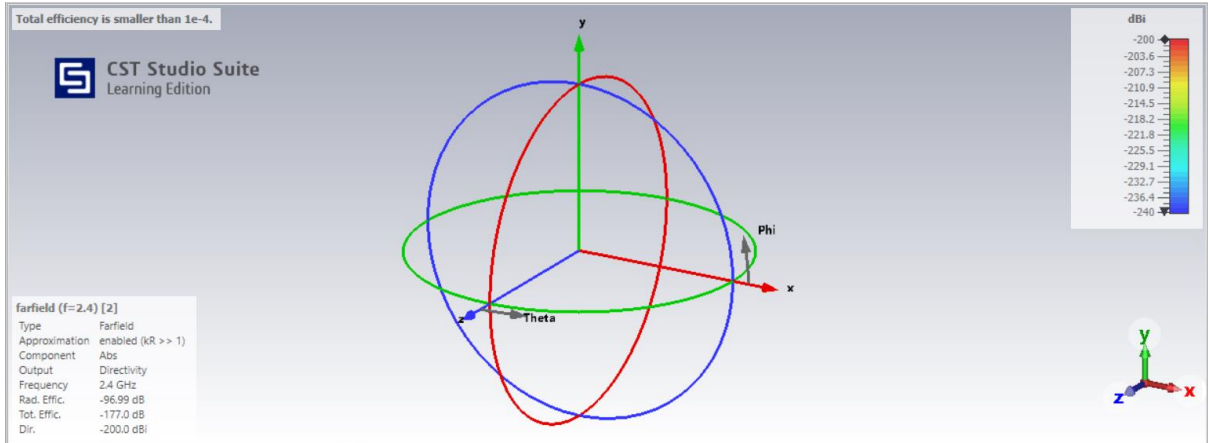


(2 numaralı porttan çıkan ve 1 numaralı porta doğru ilerleyen bir sinyalin oluşturduğu alan dağılımını gösterir.)

FARFIELD

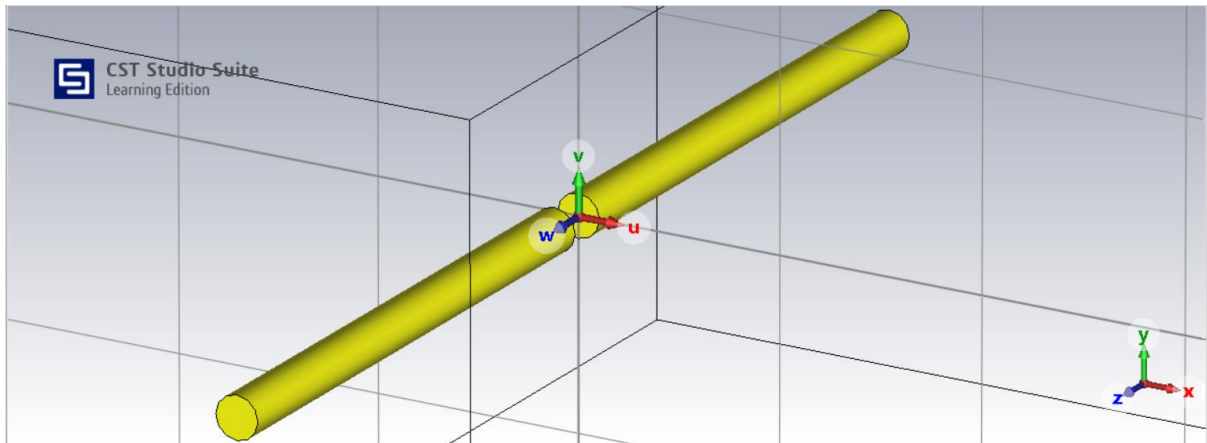


(Antenin giriş (1) portundan beslenmesiyle elde edilen farfield sonucu.)



(Antenin çıkış (2) portundan beslenmesiyle elde edilen farfield sonucu.)

DIPOLE ANTENNA



Name: WIRE

Orientation: ☐ X ☐ Y ☒ Z

Outer radius: 1.042 Inner radius: 0

Xcenter: 0 Ycenter: 0

Zmin: 0 Zmax: 27

Segments: 0

Component: component1

Material: Copper (annealed)

Name: GAP

Orientation: ☐ U ☐ V ☒ W

Outer radius: 1.042 Inner radius: 0

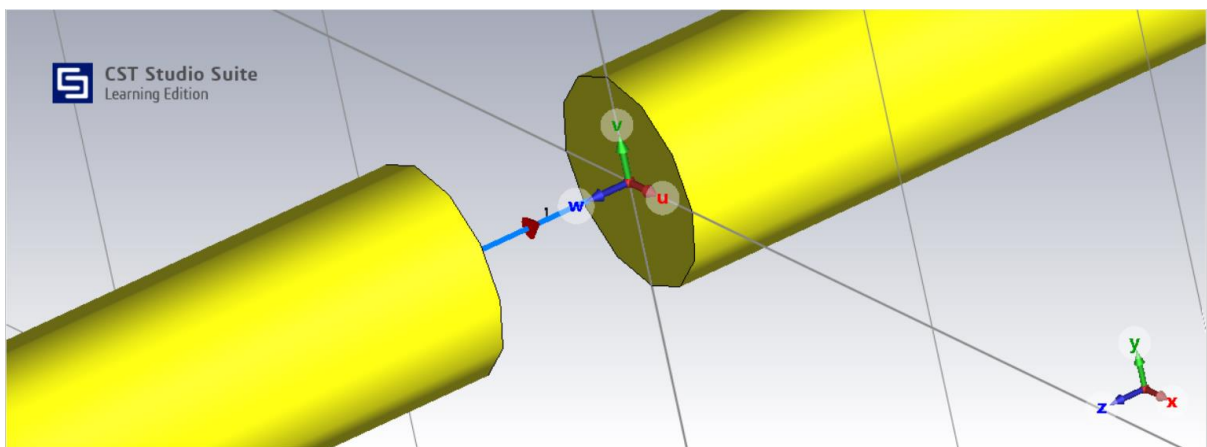
Ucenter: 0 Vcenter: 0

Wmin: -27 Wmax: 0

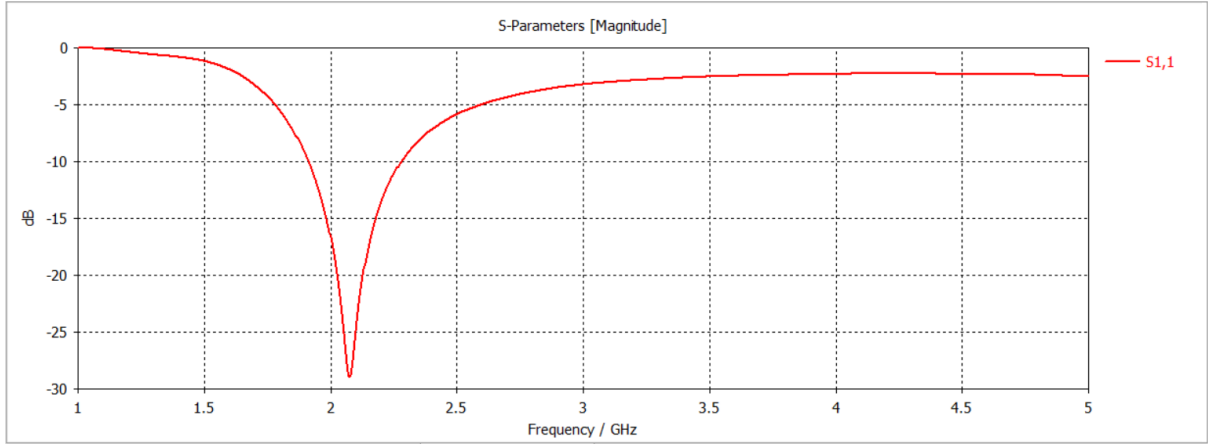
Segments: 0

Component: component1

Material: Copper (annealed)



S PARAMETER



(Yanlış uzunluk seçiminden kaynaklanan, hatalı s parametre görüntüsü)

Name: WIRE

Orientation: ☐ X ☐ Y ☒ Z

Outer radius: 1.042 Inner radius: 0

Xcenter: 0 Ycenter: 0

Zmin: 0 Zmax: 31.25

Segments: d

Component: component1

Material: Copper (annealed)

Name: GAP

Orientation: ☐ U ☐ V ☒ W

Outer radius: 1.042 Inner radius: 0

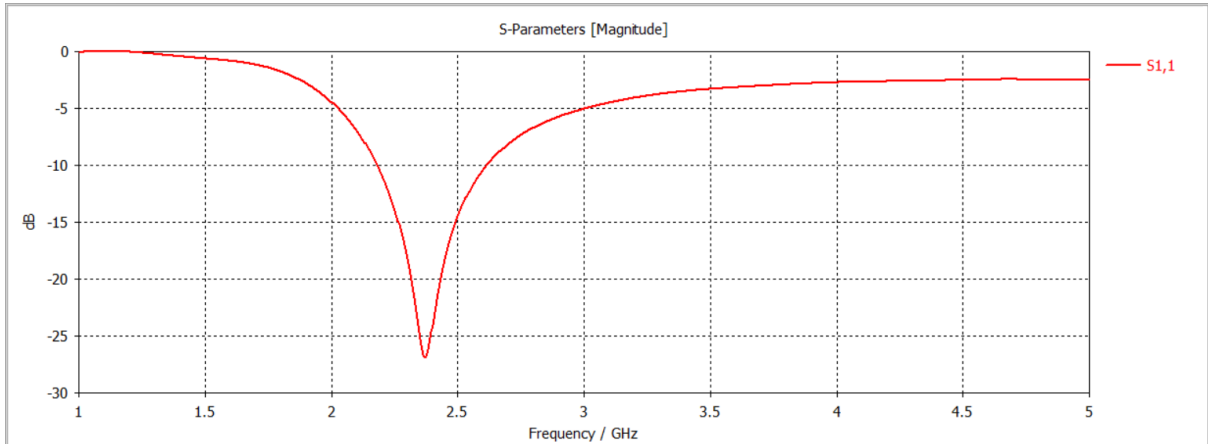
Ucenter: 0 Vcenter: 0

Wmin: -31.25 Wmax: 0

Segments: 0

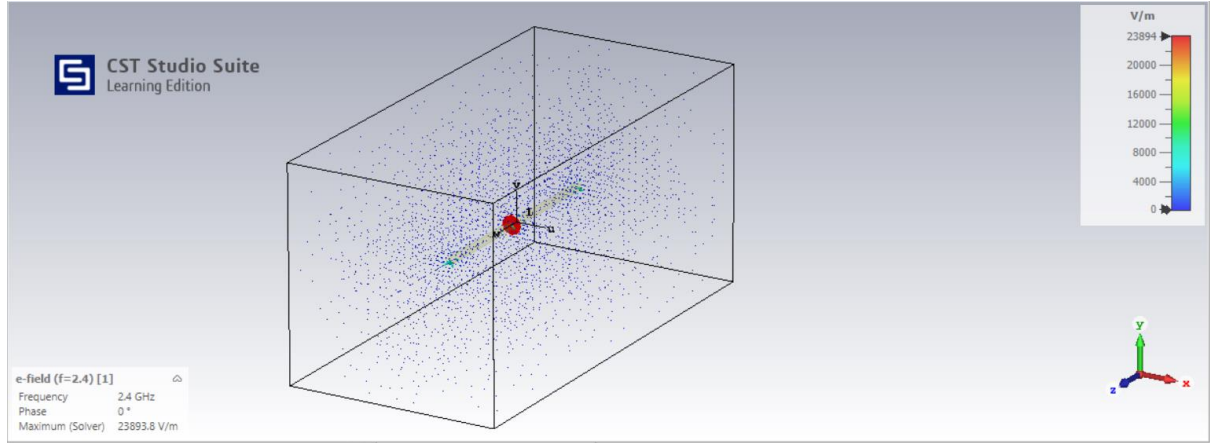
Component: component1

Material: Copper (annealed)



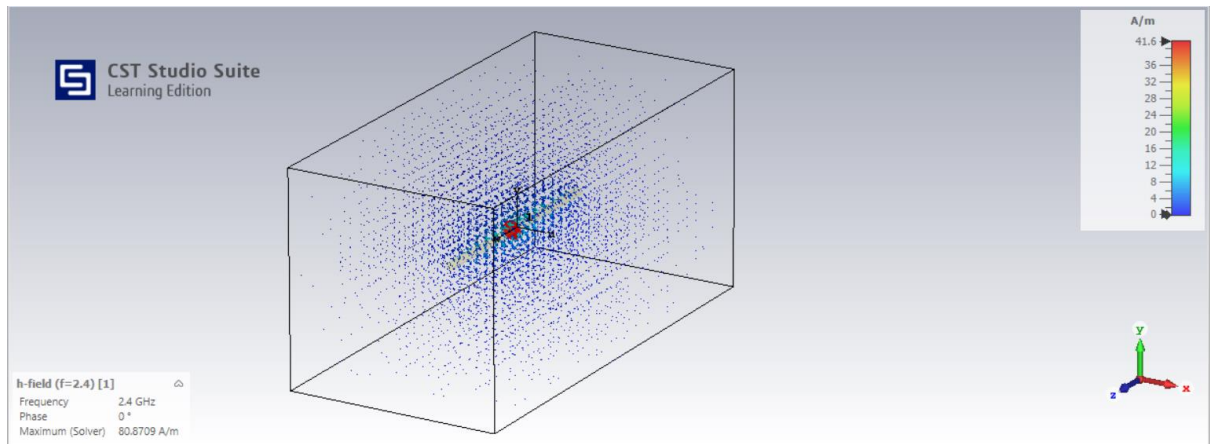
(Doğru uzunluk seçimiyle elde edilen, doğru s parametre görüntüsü)

E-FIELD



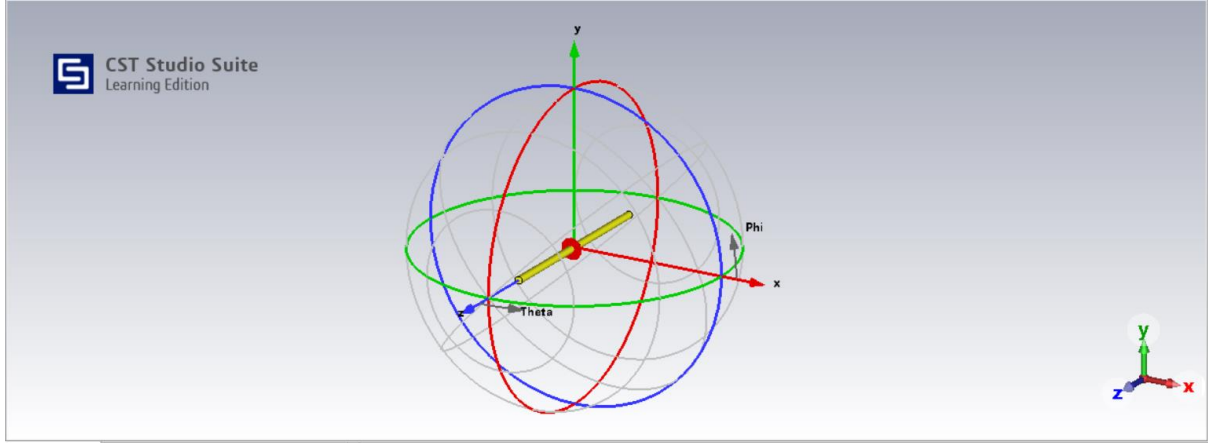
(Anten çalışırken uzayda oluşan elektrik alan vektörlerini gösterir.)

H-FIELD

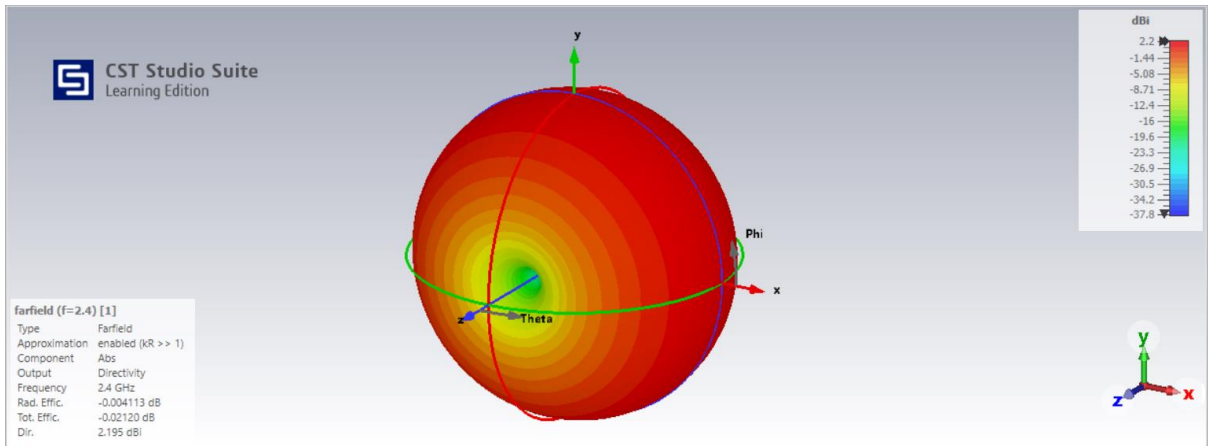


(Anten çalışırken çevresinde oluşan manyetik alan vektörlerini gösterir.)

FARFIELD & PHI-THETA

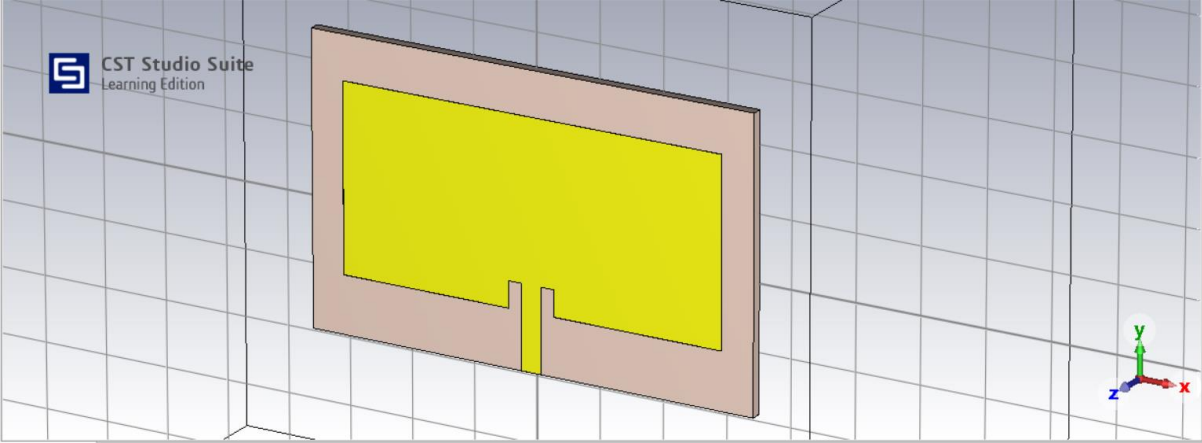


(Antenin etrafındaki koordinat sistemini ve alan analizinde kullanılan temel referans noktalarını ve yönünü gösterir.)

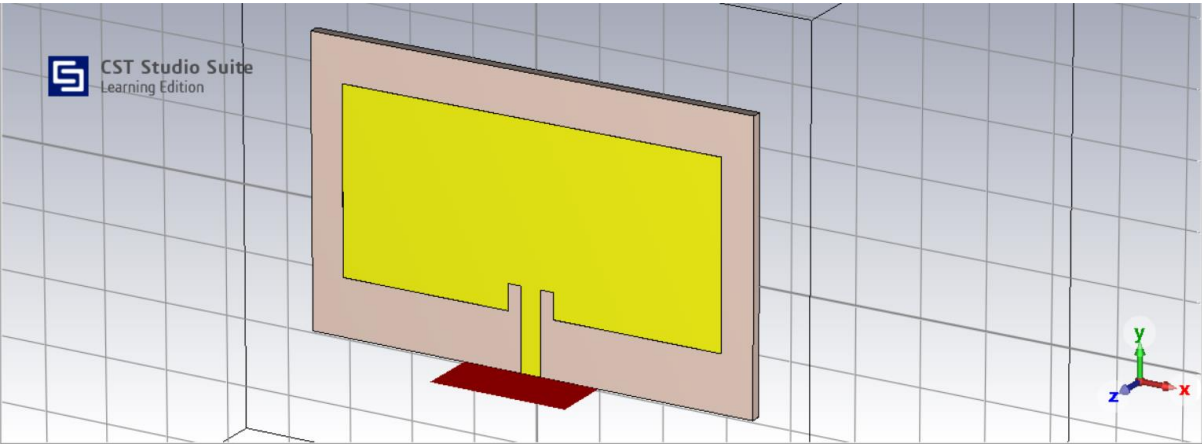


(Dipol antenin 3 boyutlu ışıma desenini ve ışıma şiddetini (kazancını) farklı yönlerde göstermektedir.)

MICROSTRIP ANTENNA



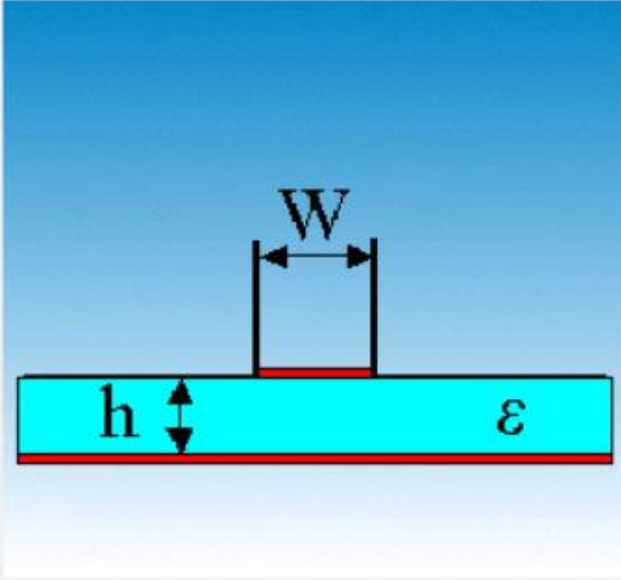
Name: dielectric		Name: ground		Name: microstrip		Name: strip	
Xmin: -23.30	Xmax: 23.30	Xmin: -23.30	Xmax: 23.30	Xmin: -18.51	Xmax: 18.51	Xmin: -1.5685	Xmax: 1.5685
Ymin: -19	Ymax: 19	Ymin: -19	Ymax: 19	Ymin: -10.2	Ymax: 18.2	Ymin: -10.2	Ymax: -19
Zmin: 0	Zmax: 1.6	Zmin: -0.035	Zmax: 0	Zmin: 1.6	Zmax: 1.635	Zmin: 1.6	Zmax: 1.635
Component: component1		Component: component1		Component: component1		Component: component1	
Material: FR-4 (lossy)		Material: Copper (annealed)		Material: Copper (annealed)		Material: Copper (annealed)	



 Impedance Calculation ×

Setup

Thin Microstrip



Length unit:

mm

Frequency:

2.4

GHz

Geometry Data

h

1.6

W

3.137

0.1

4

Line length:

9,9996e+13

Permittivity

eps_r =

4.3

☐ Include Dispersion

Impedance static

Z₀ =

50,00

Ohm

eps_{eff} =

3,27

Phase shift =

5,2100e+14

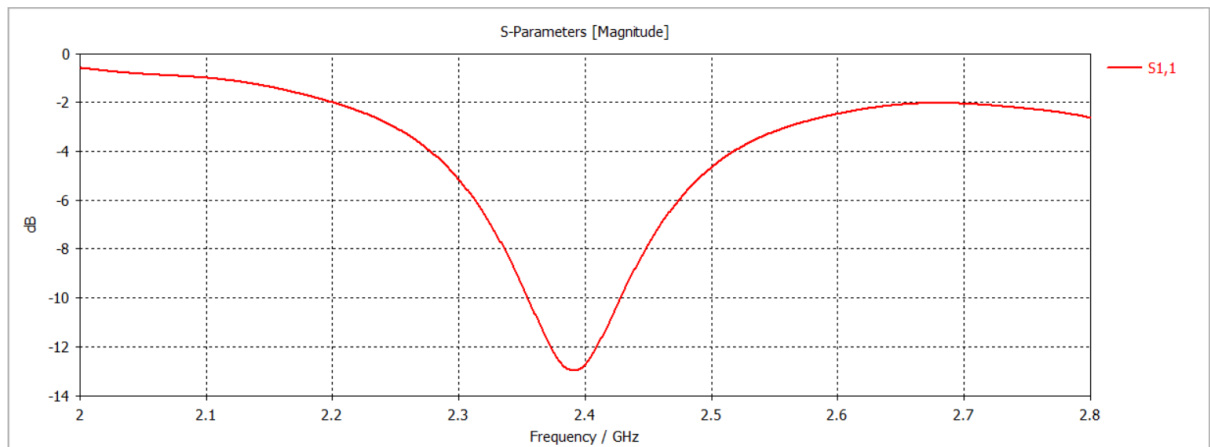
Calculate

Build 3D

Exit

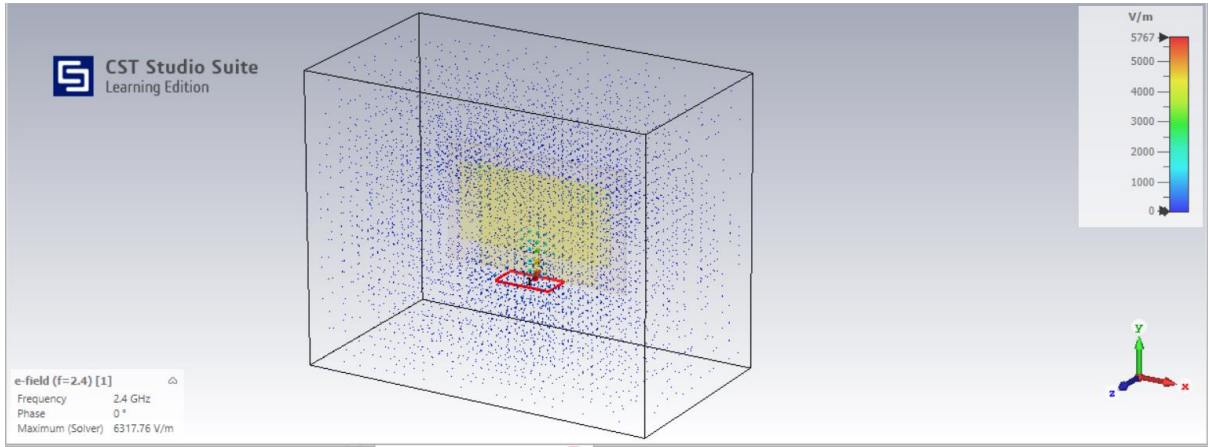
Help

S PARAMETER



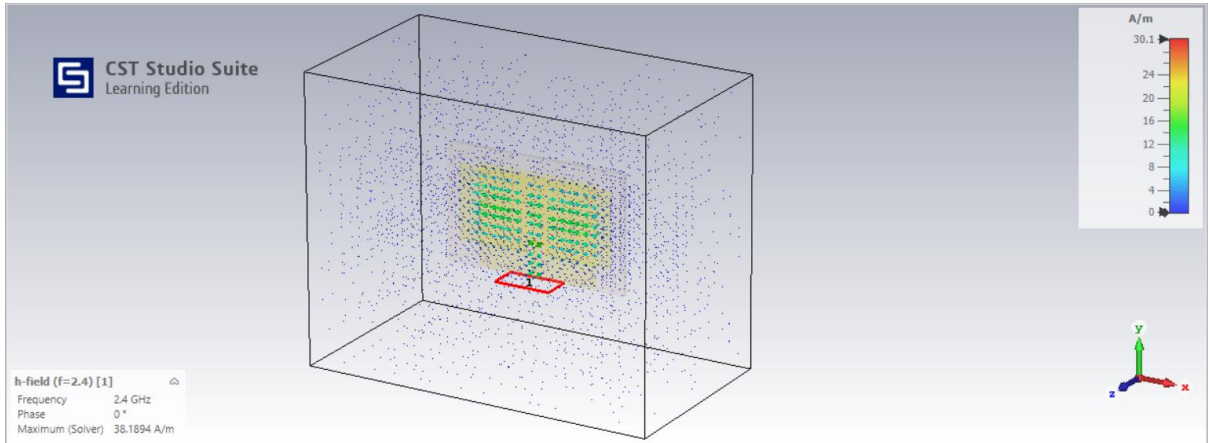
(Antenin belirli bir frekansta ne kadar gücü geri yansıttığını gösterir.)

E-FIELD



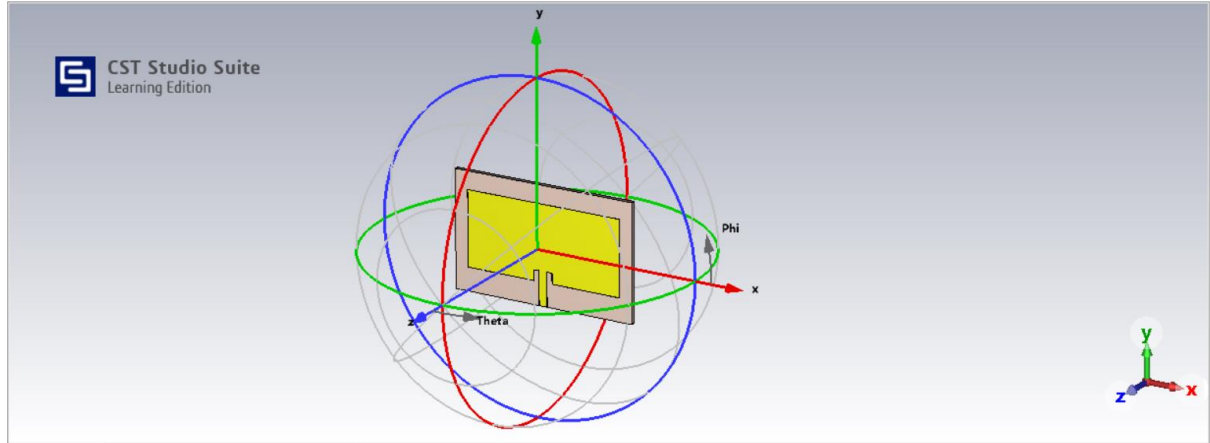
(Antenin çevresindeki elektrik alanının uzaydaki dağılımını gösterir.)

H-FIELD

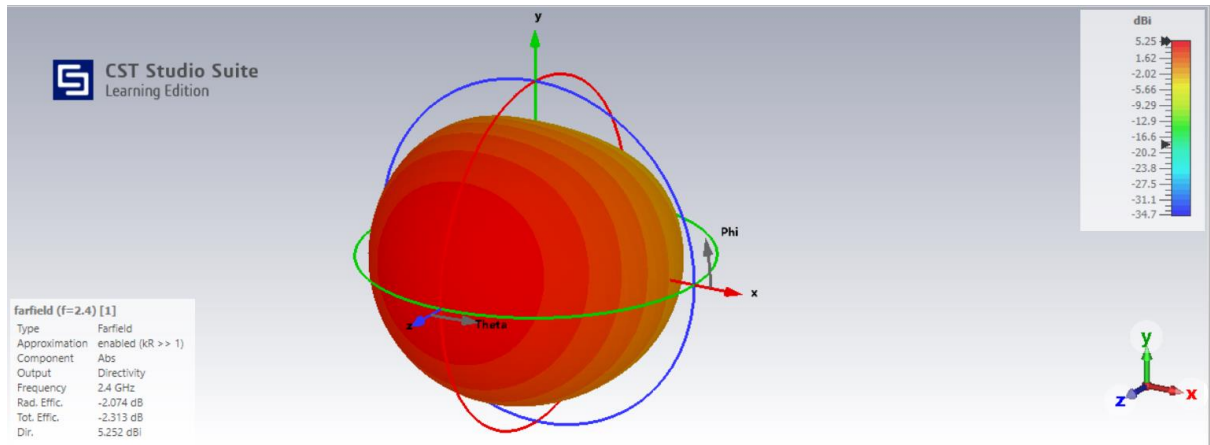


(Antenin çevresindeki manyetik alanının uzaydaki dağılımını gösterir.)

FARFIELD & PHI-THETA

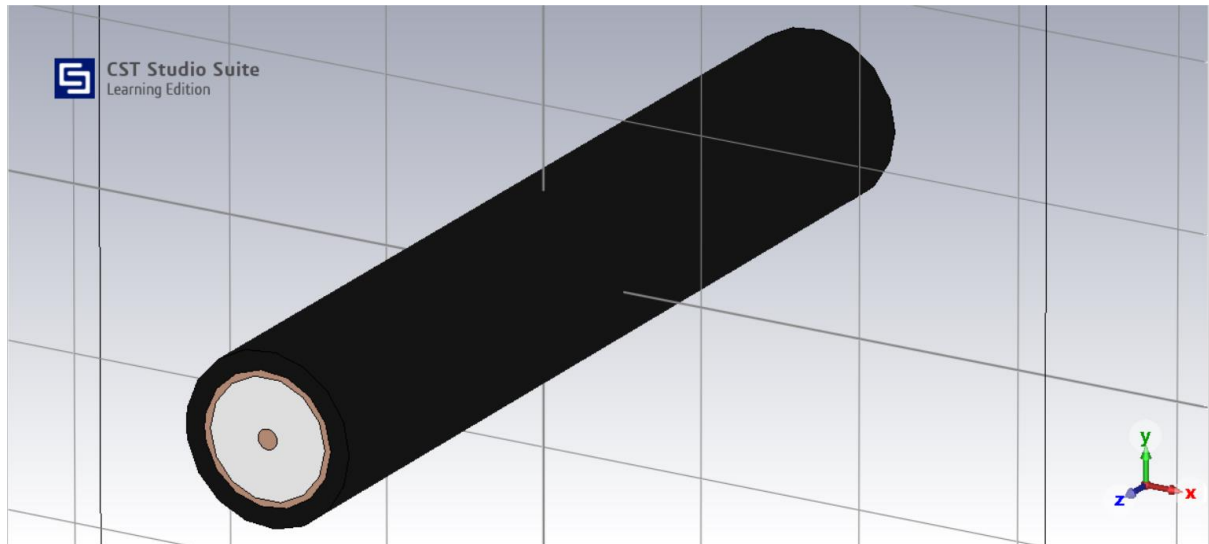


(Antenin etrafındaki koordinat sistemini ve alan analizinde kullanılan temel referans noktalarını ve yönünü gösterir.)

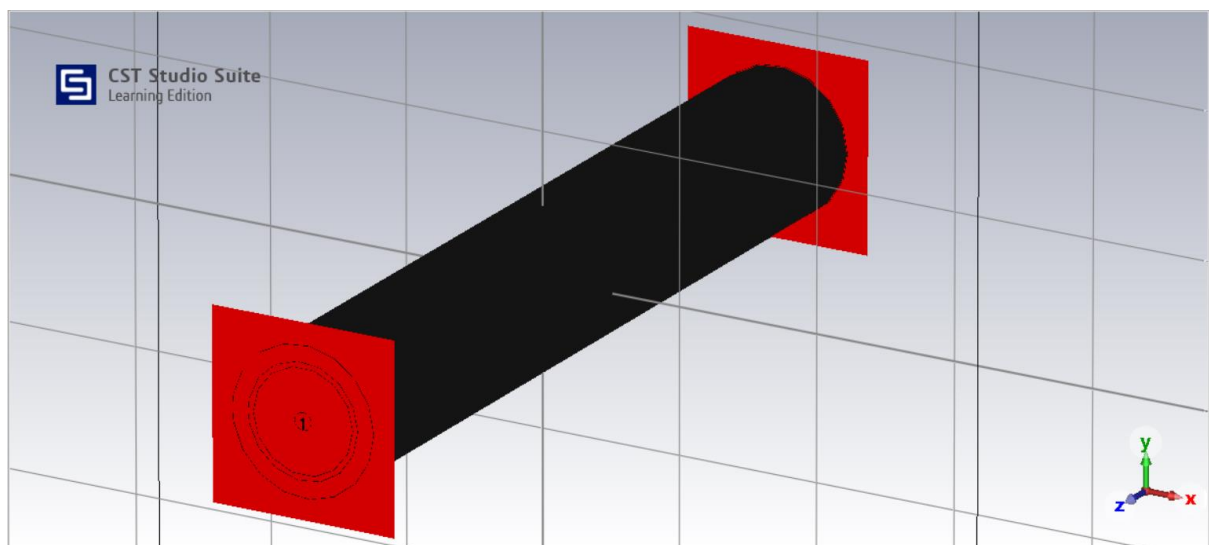


(Mikroşerit antenin 3 boyutlu ışıma desenini ve ışıma şiddetini (kazancını) farklı yönlerde göstermektedir.)

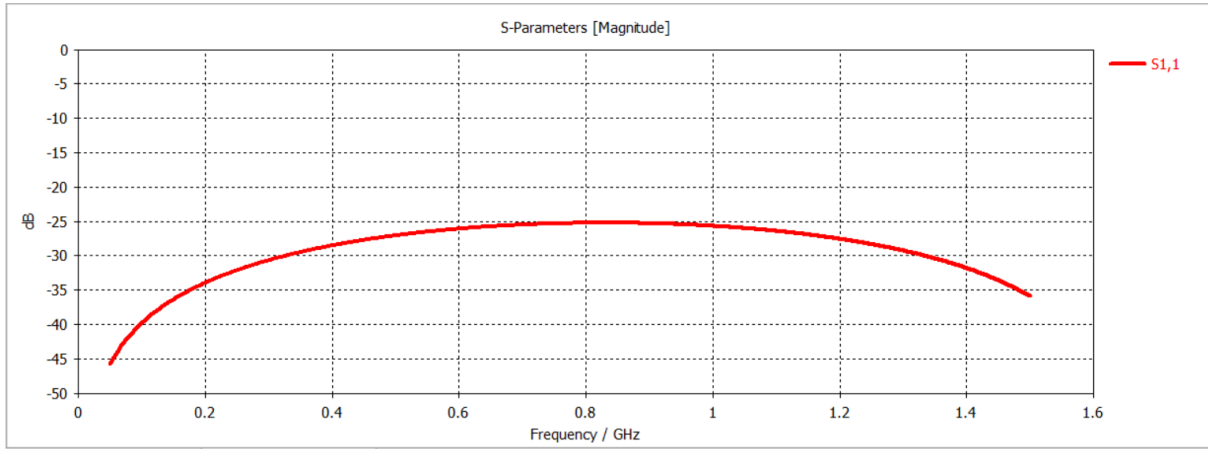
COAXIAL CABLE



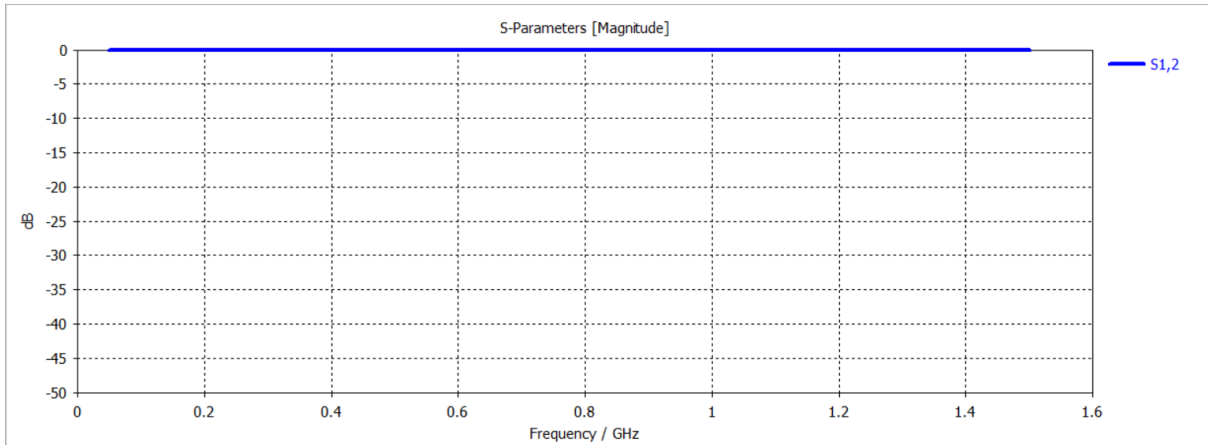
Name:	Name:	Name:	Name:
DIELECTRIC	SHIELD	INNER CONDUCTOR	JACKET
Orientation: ☐ X ☐ Y ☒ Z	Orientation: ☐ X ☐ Y ☒ Z	Orientation: ☐ X ☐ Y ☒ Z	Orientation: ☐ X ☐ Y ☒ Z
Outer radius: 3.62	Outer radius: 3.99	Outer radius: 0.61	Outer radius: 5.145
Inner radius: 0.61	Inner radius: 3.62	Inner radius: 0	Inner radius: 3.99
Xcenter: 0	Xcenter: 0	Xcenter: 0	Xcenter: 0
Ycenter: 0	Ycenter: 0	Ycenter: 0	Ycenter: 0
Zmin: -30	Zmin: -30	Zmin: -30	Zmin: -30
Zmax: 30	Zmax: 30	Zmax: 30	Zmax: 30
Segments: 0	Segments: 0	Segments: 0	Segments: 0
Component: component1	Component: component1	Component: component1	Component: component1
Material: PTFE (lossy)	Material: Copper (pure)	Material: Copper (pure)	Material: my material/PVC 1



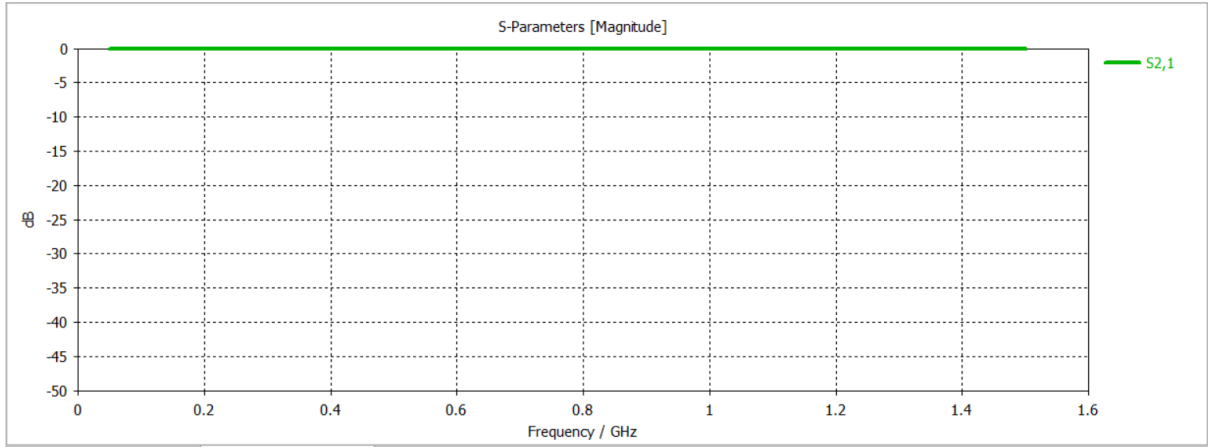
S PARAMETER



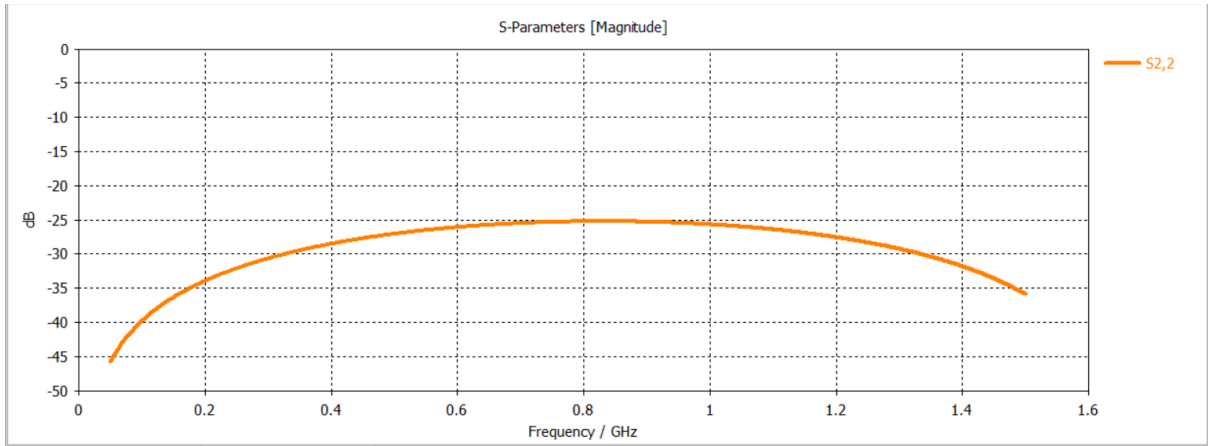
(1 numaralı porta gönderilen bir sinyalin ne kadarının geri yansıdığını gösterir.)



(1 numaralı porttan gönderilen bir sinyalin ne kadarının 2 numaralı porta iletilildiğini gösterir.)

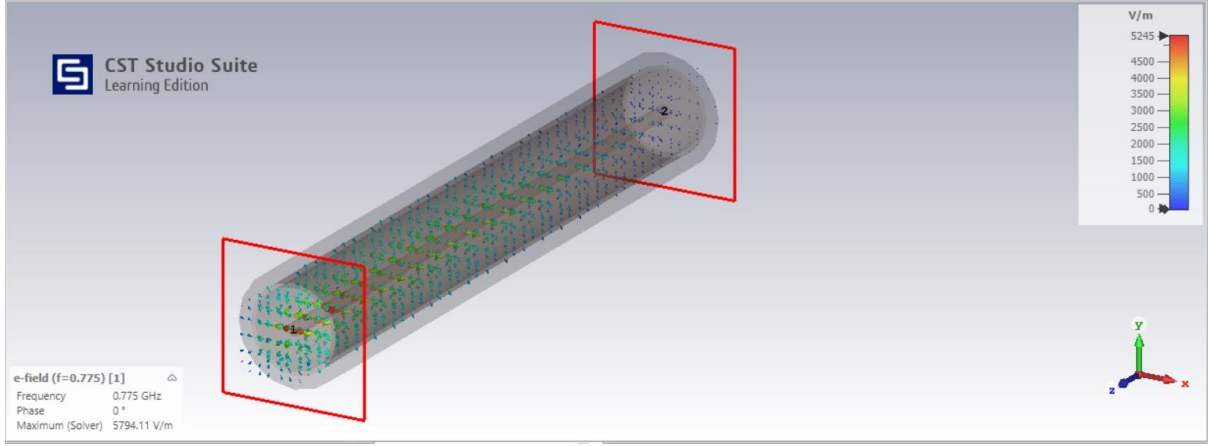


(2 numaralı porttan gönderilen bir sinyalin ne kadarının 1 numaralı porta iletiildiğini gösterir.)



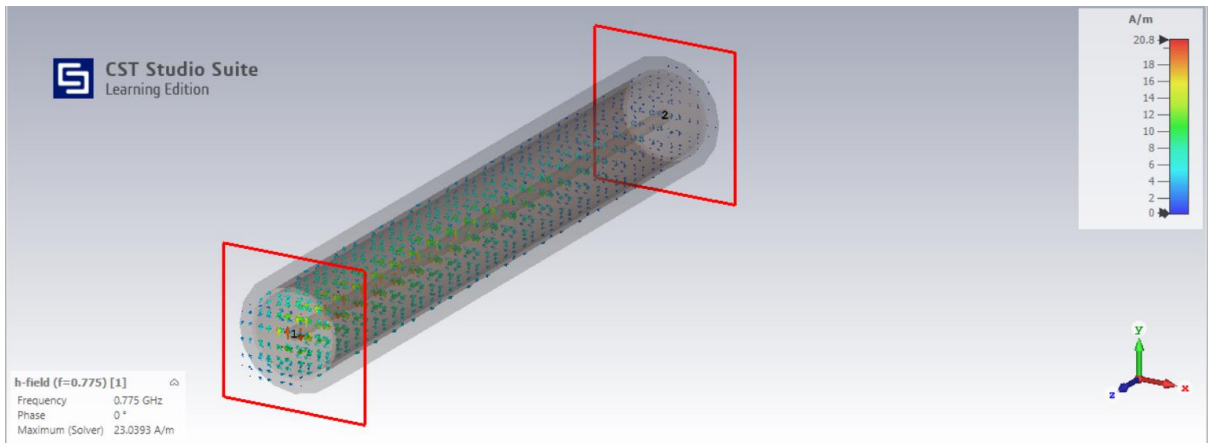
(2 numaralı porta gönderilen bir sinyalin ne kadarının geri yansıdığını gösterir.)

E-FIELD



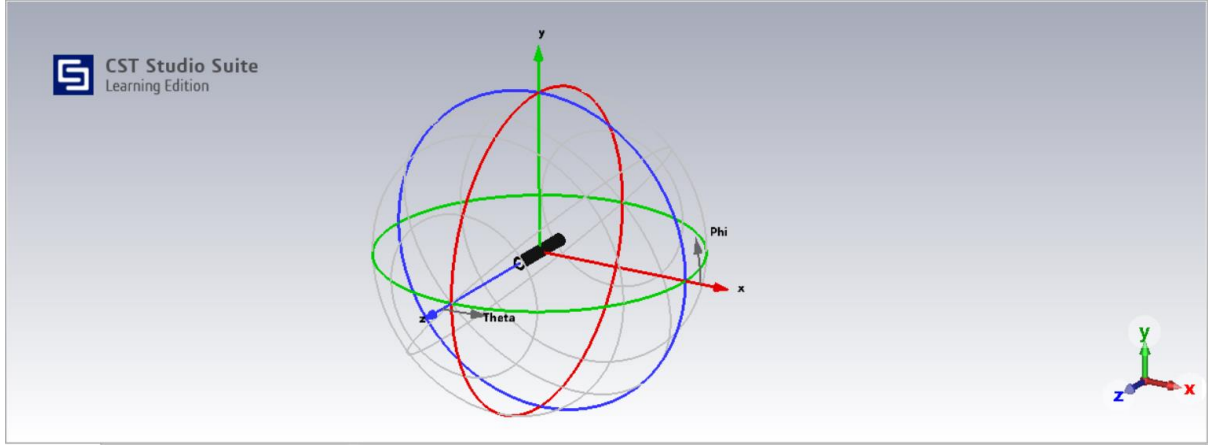
(Giriş portundan (1) gönderilen bir sinyalin dalga kılavuzunda nasıl yayıldığını ve çıkış portuna (2) nasıl ulaştığını gösterir.)

H-FIELD

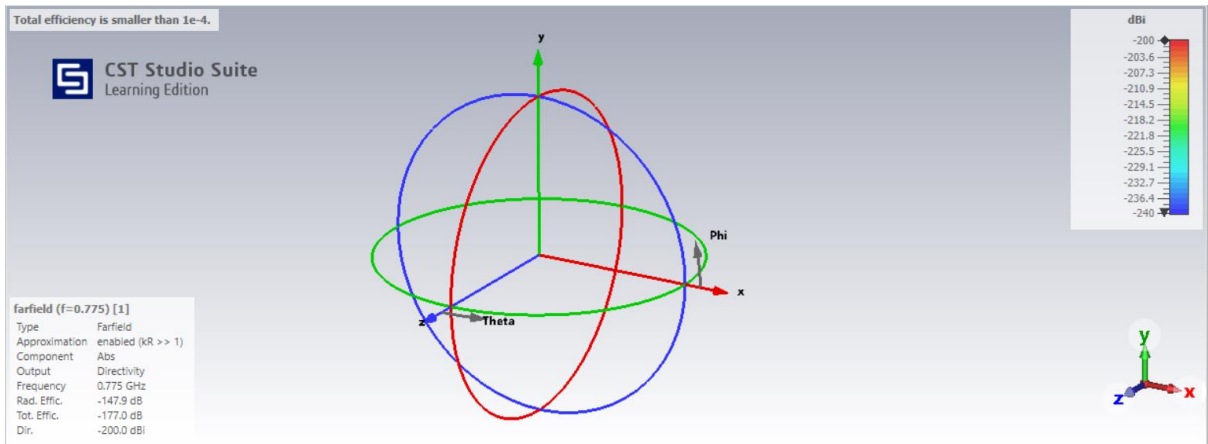


(1 numaralı porttan çıkan ve 2 numaralı porta doğru ilerleyen bir sinyalin oluşturduğu alan dağılımını gösterir.)

FARFIELD



(Antenin giriş (1) portundan beslenmesiyle elde edilen farfield sonucu.)



(Antenin çıkış (2) portundan beslenmesiyle elde edilen farfield sonucu.)

CST HESAPLAMALARI

WAVEGUIDE (WR340)

$$\lambda(c \div f) = (3 \times 10^8) \div 5 = 125$$

$$\lambda c(2a) = 2 \times 86,36 = 172,72$$

$$\lambda g = \left(\frac{\lambda}{\sqrt{1 - \left(\frac{\lambda}{\lambda c}\right)^2}} \right) = 181,133$$

$$a \text{ kenarı} = 86,36$$

$$b \text{ kenarı} = 43,18$$

$$c = 3 \times 10^8$$

$$L(2\lambda g) = 362,266$$

$$t \text{ kalınlığı} = 2,038$$

DIŞ

$$x \min \left(-\frac{a}{2} - t \right) = -45,206$$

$$y \min \left(-\frac{b}{2} - t \right) = -23,628$$

$$x \max \left(\frac{a}{2} + t \right) = 45,206$$

$$y \max \left(\frac{b}{2} + t \right) = 23,628$$

$$z \min = 0$$

$$z \max(L) = 362,266$$

İÇ

$$x \min = -43,18$$

$$y \min = -21,59$$

$$x \max = 43,18$$

$$y \max = 21,59$$

$$z \min = 0$$

$$z \max = 362,266$$

COAXIAL CABLE (RG11-75Ω)

	MATERIAL	DIAMETER
INNER CONDUCTOR	copper	1,22
DIELECTRIC	PE	7,24
SHIELD	copper	7,98
JACKET	PVC(black)	10,29

DIPOLE ANTENNA

$$\lambda = 0,125 \quad r = 1,042 \quad G(2r) = 2,048 \quad L\left(\frac{\lambda}{2}\right)$$